Chapter 1 described how to do basic mathematics with *Mathematica*. For many kinds of calculations, you will need to know nothing more. But if you do want to use more advanced mathematics, this chapter discusses how to do it in *Mathematica*.

This chapter goes through the various mathematical functions and methods that are built in to *Mathematica*. Some calculations can be done just by using these built-in mathematical capabilities. For many specific calculations, however, you will need to use application packages that have been written in *Mathematica*. These packages build on the mathematical capabilities discussed in this chapter, but add new functions for doing special kinds of calculations. Appendix A gives some examples of such packages.

Much of what is said in this chapter assumes a knowledge of mathematics at an advanced undergraduate level. If you do not understand a particular part of it, then you can probably assume that you will not need to use that part.

The emphasis in this chapter is on *what* *Mathematica* does, not *how* it does it. Except for a few cases in numerical mathematics, you should never need to know how *Mathematica* actually works inside.

About the illustration overleaf:

This is a picture of the Euler gamma function $\Gamma(z)$ in the complex plane. The height of the surface gives the absolute value of $\Gamma(z)$; the shading represents the phase of $\Gamma(z)$. You can see spikes at integer positions along the negative real axis, corresponding to poles in $\Gamma(z)$.

The actual *Mathematica* command used to produce this picture was `Plot3D[{Abs[Gamma[x + I y]], `*shading*`}, {x, -2.5, 5}, {y, -6, 1}]`, where *shading* was `GrayLevel[(Pi+Arg[Gamma[x+I y]])/(2Pi)]`. The following option settings were used: `PlotRange->{0, 6}`, `ViewPoint->{-0.8, -2, 1.2}`, `BoxRatios->{1.2, 1, 0.7}`, `PlotPoints->40`.

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